



Artificial Intelligence and Machine Learning

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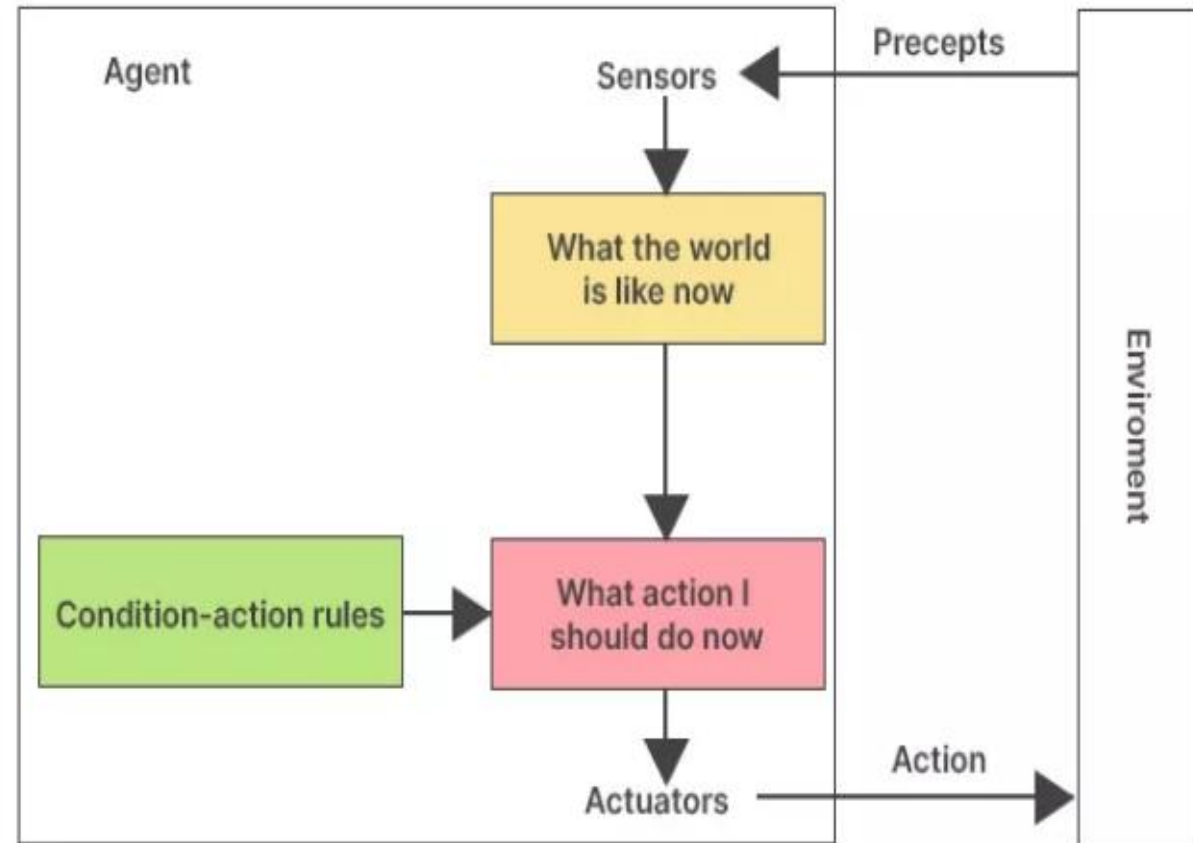
AI Landscape

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What is an Agent?

Agent: Any **entity** that perceives its environment through sensors and acts upon that environment using actuators



AIML

Intelligent Agent

AI Agent: A **computational system** that operates autonomously to achieve goals

Example: A robot vacuum cleaner senses dust (sensor) and moves to clean (actuator).



AIML

Intelligent Agent

Characteristics

Autonomy: Operates without human intervention

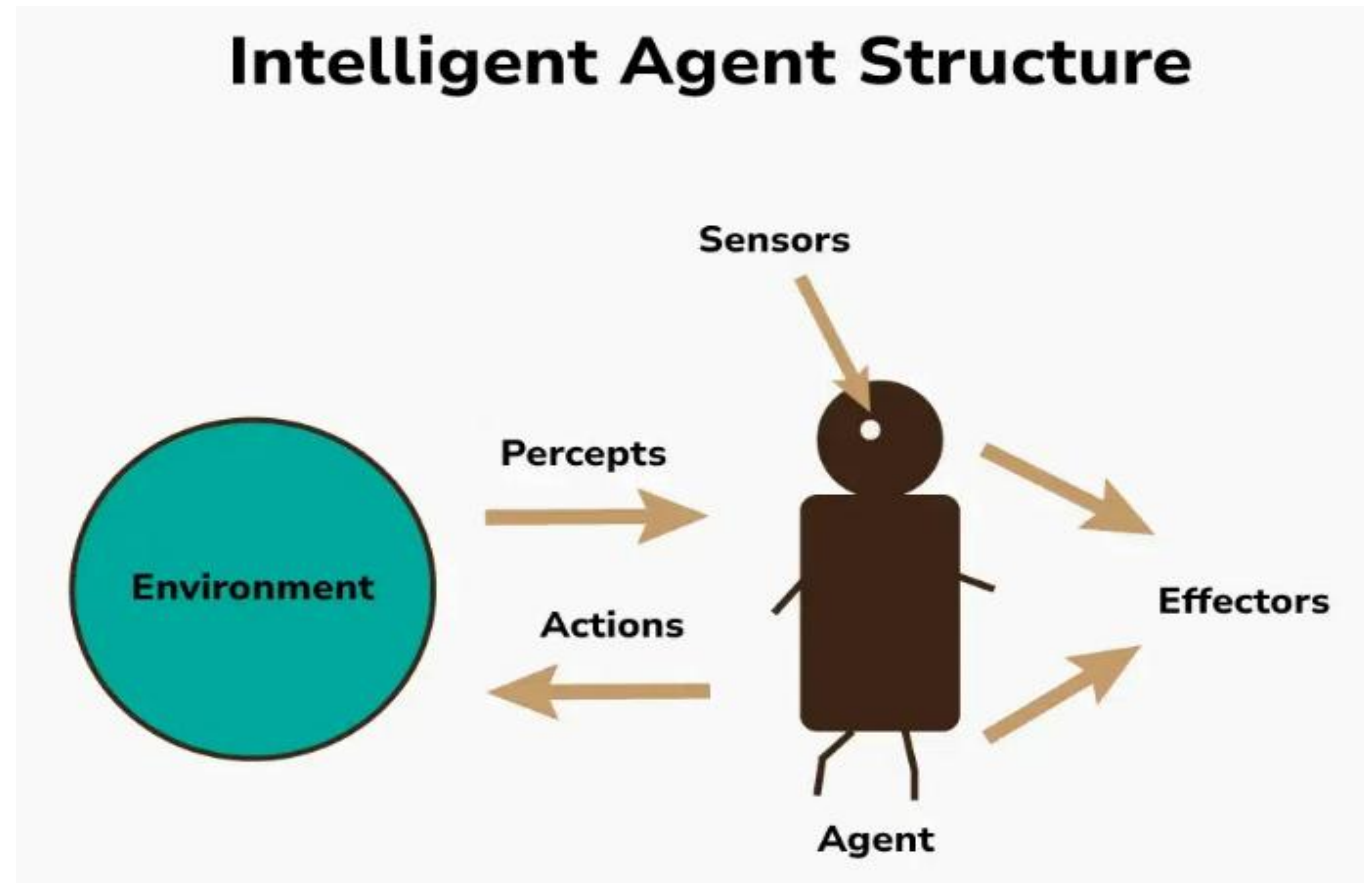
Perception: Uses sensors to observe the environment

Action: Performs tasks via actuators

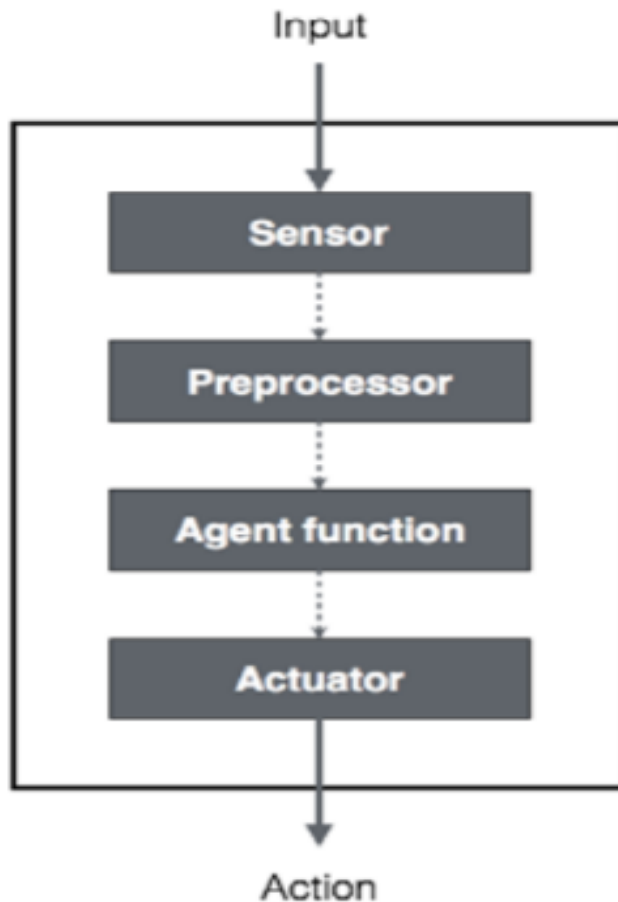
Goal-oriented: Acts to achieve specific objectives

Adaptability: May learn or adapt based on experience

Intelligent Agent: Program that can perceive its environment, make decisions, take action and perform services based on its environment, user input and previous experiences



Architecture

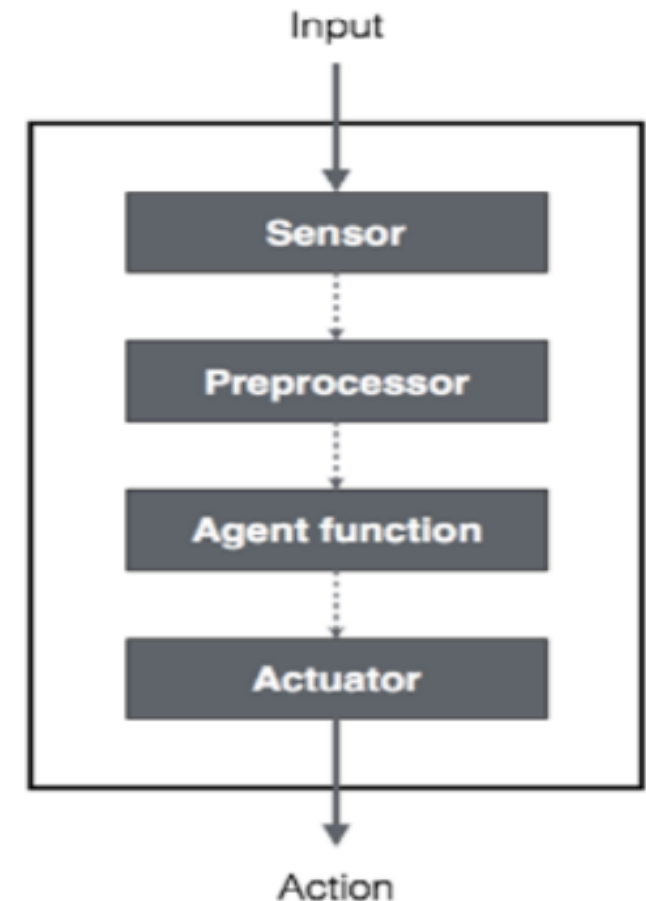


Sensors are the components that gather data from the agent's environment

Preprocessors take the raw data from sensors and convert it into a usable format.

The **Agent Function** is the core decision-making system. It uses the processed data to determine the best actions the agent should take to achieve its goals. **This involves applying algorithms, rules, or models** to make informed decisions.

Actuators are the components that execute the decisions made by the agent function. They translate the agent's decisions into physical actions



Example: Self-Driving Car Agent

- Sensors: Cameras, radar, GPS
- Actuators: Steering, brakes
- Agent Program: Detects lanes, avoids obstacles, navigates

Activity

Identify sensors, actuators and agent function for the following systems:

- Siri
- Amazon Alexa
- Netflix Recommendations
- Chess-playing bots



THANK YOU

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